

Thinking in reverse

Note Title

2016/02/21

1. Write the functions as quotient, sum or product functions.

$$y = \frac{x^2+1}{\sqrt{x}} \quad \begin{array}{l} \text{--- } f(x) \\ \text{--- } g(x) \end{array} \Rightarrow \left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)} = \frac{x^2+1}{\sqrt{x}}$$

$$y = \frac{x^2}{\sqrt{x+1}} \quad \begin{array}{l} \text{--- } f(x) \\ \text{--- } g(x) \end{array} \Rightarrow (fg)(x) = f(x) \cdot g(x) = x^2 \cdot \sqrt{x+1}$$

$$y = x + \sin x \quad \begin{array}{l} \text{--- } f(x) \\ \text{--- } g(x) \end{array} \Rightarrow (f+g)(x) = f(x) + g(x) = x + \sin x$$

2. Identify the function f , g and h if

$$\underbrace{f \circ g}(x) = \sqrt{x^2+1}$$

$\underbrace{f}_{\downarrow}(\underbrace{g}_{\downarrow}(x))$
 $\underbrace{\quad}_{\downarrow} \quad \underbrace{\quad}_{\downarrow}$
 $\sqrt{\quad} \quad x^2+1$

$$f(x) = \sqrt{x}$$

$$g(x) = x^2+1$$

$$f(g(x)) = \sqrt{x^2+1}$$

$$f \circ g \circ h(x) = \frac{1}{\sqrt{x+1}}$$

$$\downarrow$$

$$f(g(h(x)))$$

$$h(x) = x+1$$

$$g(x) = \sqrt{x}$$

$$f(x) = \frac{1}{x}$$

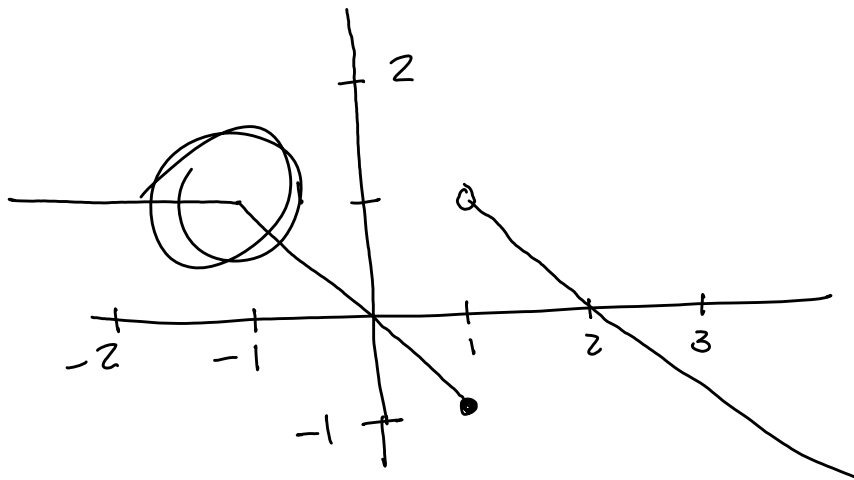
$$f \circ g \circ h(x) = (\sqrt{x} + 1)^3$$

$$h(x) = \sqrt{x}$$

$$g(x) = x+1$$

$$f(x) = x^3$$

4. Give the formula of the function:



$$\Rightarrow y = \begin{cases} 2-x & \text{if } x > 1 \\ -x & \text{if } -1 \leq x \leq 1 \\ 1 & \text{if } x < -1. \end{cases}$$